National University of Computer and Emerging Sciences



Laboratory Manuals

for

Computer Networks - Lab

(CL -3001)

Course Instructor	Dr. Abdul Qadeer
Lab Instructor	Ms. Halima Sadia
Section	BCS-5A
Semester	Fall 2026

Department of Computer Science
FAST-NU, Lahore, Pakistan

Lab Manual 05

CLOs

- CLO 2: Demonstrate an understanding of the fundamental concepts of socket programming, types of protocols, and IP addressing.

Objectives

By the end of this lab, students should be able to:

- Implement a basic TCP client-server application using Python sockets.
- Understanding connection-oriented client-server communication
- Handling multiple clients using a persistent TCP server

Prerequisites

Students should have basic knowledge of:

- Python functions, strings, lists, and conditional statements.
- Client-server communication.
- TCP and transport-layer protocols.
- Basic IP addressing.

TCP (Transmission Control Protocol)

TCP is a connection-oriented transport layer protocol that provides reliable and ordered delivery of data between applications. Before data is exchanged, the client and server establish a connection.

Characteristics of TCP

- TCP is connection-oriented and requires a connection to be established before communication.
- TCP provides reliable and ordered delivery of data.
- TCP ensures that lost or corrupted data is retransmitted.
- A TCP server can continue running and serve multiple clients one after another.

In-Lab Statement 1 [5 Marks]

Write a TCP client and server program that can communicate with each other using the following messages:

- Client sends: "Hello I am client and My ID is 1"
- Server responds: "Hello I am server. Your received ID is 1"

Requirements:

- The client ID must be a single digit from 0 to 9.
- Run the client and server on the same machine.
- The server must remain running continuously and should not terminate after serving one client.

- Multiple clients should be able to connect to the server one by one.
- The server should extract the client ID from the received message and include it in the response.

Sample Test Bench:

Client 1 sends:

```
Hello I am client and My ID is 1
```

Client 2 sends:

```
Hello I am client and My ID is 2
```

Server response to Client 1:

```
Hello I am server. Your received ID is 1
```

Server response to Client 2:

```
Hello I am server. Your received ID is 2
```

In-Lab Statement 2 [15 Marks]

Write a TCP client and server program in which the client sends a sentence to the server.

Requirements:

- The server must receive the complete sentence from the client.
- The server must display the received sentence.
- The server must reverse every word that contains one or more vowels.
- Words containing no vowels should remain unchanged at the server.
- The server must send the resulting sentence back to the client.
- The client must then reverse the words that contain no vowels.
- The client must display the final resulting sentence.
- The server must remain running continuously and serve multiple clients one by one.

Sample Test Bench:

Client sends to server:

```
the birds fly in dry sky at night
```

Server displays and returns:

```
eht sdrib fly ni dry sky ta thgin
```

Client displays:

```
eht sdrib ylf ni yrd yks ta thgin
```

Submission Requirements

Submit the following files:

In-Lab Statement 1:

- TCP_Client.py
- TCP_Server.py

In-Lab Statement 2:

- TCP_Client.py
- TCP_Server.py

Screenshots showing the execution of both in-lab statements.

Important Notes

- Use TCP sockets for both statements.
- The server must be started before the client.
- The server must remain running after serving each client.
- Clients will connect to the server **one at a time**.
- The client ID in Lab Statement 1 must be a single digit from **0 to 9**.
- For Lab Statement 2, the vowels are a, e, i, o, u, and vowel checking should be case-insensitive.
- Preserve the spaces between words when processing the sentence.
- Do not change the required output format.
- The server must continue waiting for another client after completing a request.
- Terminate the server only intentionally when testing is complete.